Total Points: [10] [U1HW7 - W10 - Paddle Boat 2]

1. For a boat to move forward, the force of propulsion must be:

- Equal to the force of gravity
- Zero
- Weaker than the force of air resistance
- Directed upward
- Directed downward
- Stronger than the force of water resistance (drag)

Answer:

2. Why does running in a swimming pool feel like moving in "slow motion"?

- Your eyes are closed
- The water temperature relaxes your muscles too much
- The light refracts differently
- The water exerts a strong resistance force against your movement
- The water makes you lighter
- The pool floor is slippery

Answer:

3. To maximize the speed of a paddle boat, how should the paddle blades interact with the water?

- They should barely touch the surface to avoid wetness
- They should be made of heavy lead
- They should be completely round and smooth to slip through
- They should remain stationary
- They should move sideways like a fish tail
- They should have enough surface area to push a large volume of water backward

Answer:

4. When constructing the axle for a paddle boat, why must it be able to rotate freely?

- To generate electricity
- To allow the rubber band to hold the boat together
- To increase the weight of the boat
- To make the boat look complicated
- To ensure the boat sinks
- To allow the release of stored energy without excessive friction

Answer:

• Inside the sealed cabin

Answer:

5. During the "Evaluate and Revise" step of the design process, what should you do if your boat sinks?

• Wait for the water to evaporate

• Paint it a brighter color

• Analyze why it sank (e.g., leaks, weight) and modify the design

• Claim that it is a submarine and stop working

• Hide the project

• Blame the water for being too wet

Answer:

6. If you are building a paddlewheel boat, where is the most effective place to mount the paddlewheel to ensure forward motion?

• On the rear (stern) or the sides, partially submerged

• Dragging 3 feet behind on a string

• Underneath the bottom, fully enclosed

• On the front (bow) facing upward

• On top of the roof, spinning in the air

7. If you want to reduce drag (water resistance) on your boat, what shape should the front (bow) generally be?

• A pointed or V-shape (streamlined)

• A sphere

• A vertical wall

• A hollow ring

• A jagged, rough shape

• A flat, wide square

Answer:

8. Why does a swimmer (or a boat) move slower in water than a runner moves on land, assuming the same effort?

• Water is denser and provides more resistance than air

• Gravity is stronger in water

• The swimmer is usually tired

• Water has no friction

• The runner has better shoes

• Air pushes the runner forward

Answer:

9. Newton's Third Law implies that for every action, there is an equal and opposite reaction. How does this apply to a paddle boat?

• The weight of the boat creates a reaction of sinking

• The gravity pulls the boat left

• The action of pushing water back creates the reaction of moving the boat forward

• The friction of the water stops the reaction

• The boat pushes air up to move down

• The boat pulls the water forward to move forward

Answer:

10. How does the wing movement of a penguin under water relate to engineering a boat?

• Penguins use their wings as paddles to push water backward

• Penguins use magnetic fields to navigate

• Penguins use jet propulsion

• Penguins walk on the ocean floor

• Penguins spin in spirals to create a vortex

• Penguins inflate their bodies to float

Answer: